

# 1. PillVision: D0 architecture and design

System decomposition, interfaces, information flow and design rationale

## Goal and system boundary

Estimate pill count on a camera-visible tray, including touching or partially overlapping pills, and flag uncertain results for operator review. The basic input is a batch of pills placed on the counting tray and a start/reset command from the operator. The output is a displayed count (maybe with confidence) and session status; when observations are insufficient, the output requests operator review instead of asserting an exact count.

## Reading the diagrams

The block diagram shows components and their interfaces; the data-flow diagram shows processing stages, transfers and the retry loop. C1–C8 identify components; I1–I13 identify connections defined in Part 4. Repeated component IDs in the data flow denote multiple stages of one component.

| Diagram element                     | Meaning                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solid blue box                      | Team software or integrated assembly.                                                                                                                                   |
| Dashed hardware box                 | Supplied hardware or a physical stage; gray in data flow, orange for the external D0 camera.                                                                            |
| Green result box                    | Operator-facing result display.                                                                                                                                         |
| Blue arrow                          | Data/control from sender to receiver, including electrical commands in data flow.                                                                                       |
| Physical coupling arrow             | Optical/mechanical transfer: dashed brown in data flow; dashed green optical or dashed brown mechanical arrows in D0. Electrical commands use solid brown arrows in D0. |
| C# / I#                             | Component ID / interface ID, matching the responsibility and interface tables.                                                                                          |
| Edges, leaders and dimension arrows | Physical outlines, part annotations and illustrative dimensions in the assembly drawings.                                                                               |

Each figure repeats its own conventions. C8 may run locally or call an external API; the laptop and MCU are deployment platforms rather than extra counting stages.

## Design basis

Component ownership and the I1–I13 logical connections extend the supplied Part 3 and Part 4 tables with C8 and I12/I13. The concrete encodings, protocol proposals and rationale below extend those tables into a consistent D0 design. They are proposed implementation choices rather than already measured or implemented behavior. Overall/tray dimensions remain unselected; physical illustrations retain illustrative geometry.

## Note on alternatives

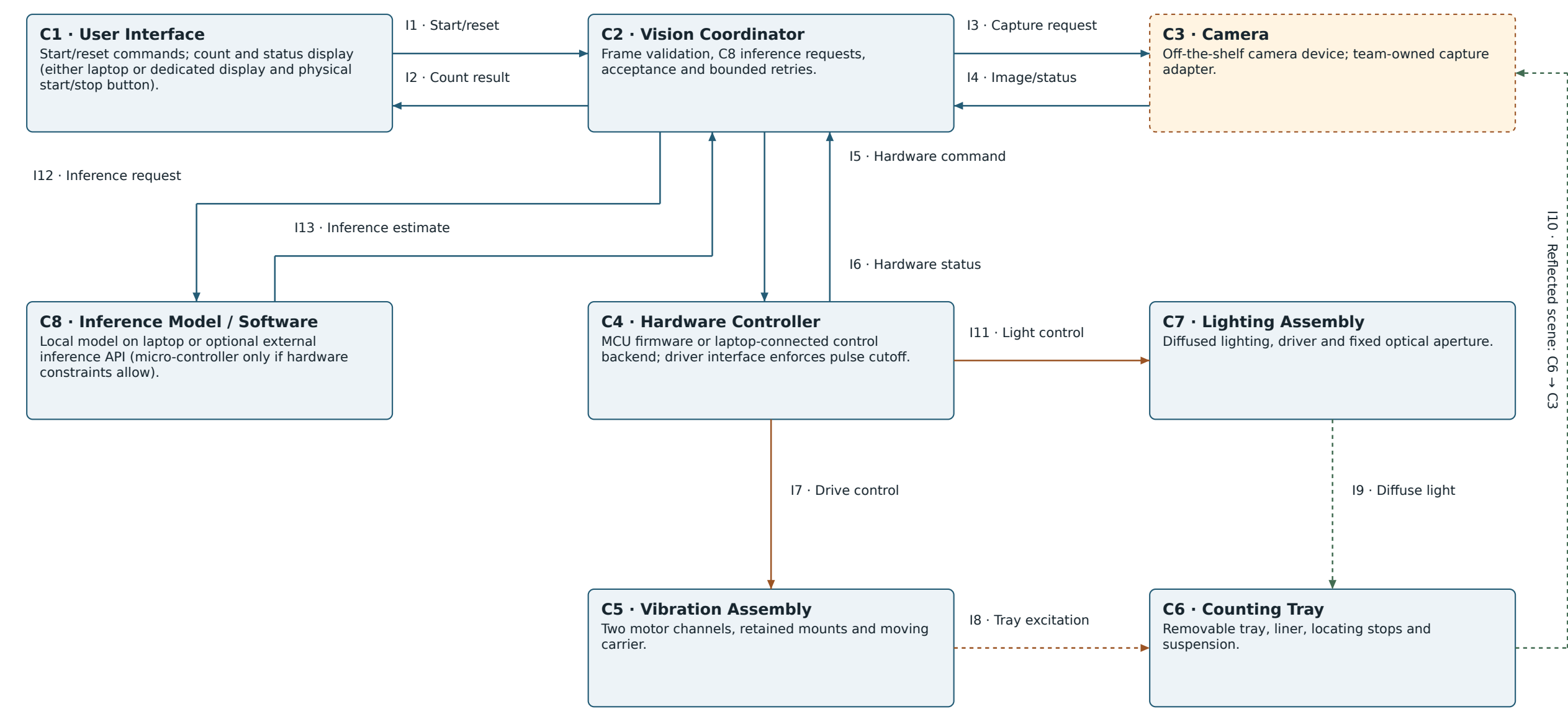
The current proposal runs UI, coordination and inference locally on a laptop, with an MCU enforcing bounded hardware actuation. A laptop-only software deployment still needs a motor/lighting controller with an independent cutoff. An MCU-led deployment may call external inference through a network-capable host or gateway; MCU memory and connectivity feasibility must be checked before choosing it. An external API adds connectivity, provider availability and image-upload latency constraints. None of these alternatives changes the component’s (C8) logical responsibility or C2’s ownership of acceptance and retries.

## 2. D0 system block diagram

Eight components; thirteen directed interfaces

**PillVision · D0 system block diagram: eight components with the I1-I13 interface contract.**

**Goal:** Estimate pill count on a camera-visible tray, including touching or partially overlapping pills, and flag uncertain results for operator review.



**Legend:** Solid blue boxes are team software or integrated assemblies; the dashed orange C3 box denotes the supplied camera plus its integration adapter. C4 may run on an MCU or laptop-connected controller; C8 may run locally or use an external API. Every routed connector starts at its sender and ends with an arrowhead at its receiver. Solid blue lines carry data/control; solid brown lines carry electrical commands; dashed brown lines carry mechanical excitation; dashed green lines carry optical coupling. I# labels match the interface table.

Figure 1: D0 system block diagram: eight components with the I1-I13 interface contract.

### Component boundaries

C1 handles the operator session; C2 coordinates observations, acceptance and retries; C8 performs inference through I12/I13. C3 converts optical observations into image buffers. C4 enforces hardware-command bounds through an MCU or a laptop-connected control interface. C5 excites C6, while C7 illuminates C6; C6 returns the reflected scene to C3.

The motor driver belongs to C5 and the lighting driver belongs to C7. The laptop can host C1, C2 and C8, and optionally the C4 backend; the camera adapter belongs to C3 integration. This is a logical component diagram rather than one box per circuit board or mechanical part.

The next page shows how this decomposition becomes an assembled device: C3 and C7 share the stationary bridge, C5 drives the isolated C6 carrier, and the base can house C4 electronics, while a laptop may remain outside the chassis.

# System architecture: assembled concept

D0 physical realization · C3–C7

## Logical components in the device

C3 camera and C7 lighting share the stationary bridge above C6, the counting tray. C5 acts on the tray carrier; the base may contain the C4 controller. A laptop outside the chassis may host C1/C2/C8 and optionally the C4 control backend.

## Structural separation

The camera remains stationary while the tray can move. This supports D3 and the I8 mechanical-excitation / I10 optical-return paths without adding a logical chassis component.

## PillVision · Assembled physical concept.

**Goal:** Estimate pill count on a camera-visible tray, including touching or partially overlapping pills, and flag uncertain results for operator review.

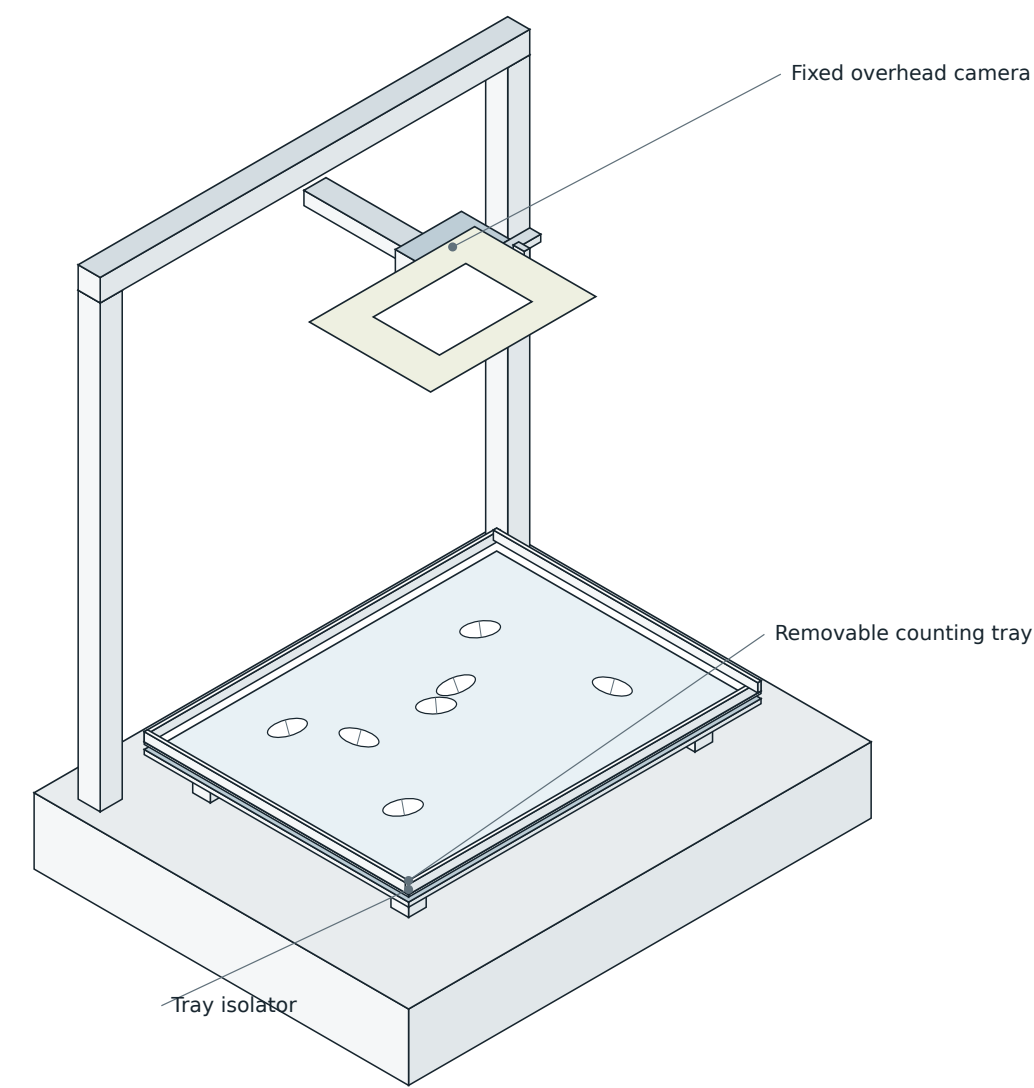


Figure 2: Assembled physical concept.

### 3. Component responsibility table

Primary owners extended from Part 3 for the new inference component

| Component                     | Responsibility                                                                                 | Interfaces in   | Interfaces out  | Primary owner    |
|-------------------------------|------------------------------------------------------------------------------------------------|-----------------|-----------------|------------------|
| C1 User Interface             | Presents the counting session to the operator.                                                 | I2              | I1              | Nekruz Ashrapov  |
| C2 Vision Coordinator         | Coordinates the counting workflow from validated observations to an accepted result or review. | I1, I4, I6, I13 | I2, I3, I5, I12 | Daniil Goncharuk |
| C3 Camera                     | Captures tray images requested by the vision processor.                                        | I3, I10         | I4              | Max Shapiro      |
| C4 Hardware Controller        | Executes bounded motor and lighting commands independently of inference.                       | I5              | I6, I7, I11     | Max Shapiro      |
| C5 Vibration Assembly         | Excites the tray to reduce severe pill clustering.                                             | I7              | I8              | Max Shapiro      |
| C6 Counting Tray              | Contains the pills in the camera-visible counting region.                                      | I8, I9          | I10             | Daniil Goncharuk |
| C7 Lighting Assembly          | Illuminates the tray consistently for image acquisition.                                       | I11             | I9              | Daniil Goncharuk |
| C8 Inference Model / Software | Produces pill-count estimates from validated images.                                           | I12             | I13             | Max Shapiro      |

#### Collaboration and task ownership

Daniil Goncharuk is primary for C2, C6 and C7; Nekruz Ashrapov for C1; Max Shapiro for C3, C4, C5 and the newly proposed C8. As recorded in Part 3, Nekruz and Max support C2, and Daniil supports C5. These are component-task owners, not permanent team specializations: the team contract specifies shared or rotating coordination, development, documentation and testing roles.

# 4. Interface specification table

High-level contracts for the thirteen D0 connections

Each row names the sender’s output and the receiver’s input for the same transfer. Record fields describe the information crossing the boundary, not a wire encoding or firmware implementation. Physical connections use mechanical or optical types rather than software messages.

| ID  | Connection                  | Named inputs / outputs and types                                                                                                                                                                                             | Format                            | Protocol / connection method                  | On error                                                                    | Handled by                                                |
|-----|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------|
| I1  | C1 → C2<br>Start/reset      | <b>Out:</b> command: SessionCommand<br><b>In:</b> session_request: SessionCommand<br>action: enum(start, reset)                                                                                                              | Structured command                | Internal application event                    | Reject invalid actions and display an error.                                | C2 validates; C1 displays.                                |
| I2  | C2 → C1<br>Count result     | <b>Out:</b> result: CountResult<br><b>In:</b> display_result: CountResult<br>count: nonnegative integer or null; confidence: real in 0-1 or null; status: enum(complete, review, error)                                      | Structured result                 | Internal application event                    | Display review/error rather than an unverified count.                       | C2 determines status; C1 displays.                        |
| I3  | C2 → C3<br>Capture request  | <b>Out:</b> request: CaptureCommand<br><b>In:</b> capture_request: CaptureCommand<br>action: capture                                                                                                                         | Structured capture command        | Camera capture API                            | Stop the counting cycle and report capture failure.                         | C3 reports failure; C2 handles it.                        |
| I4  | C3 → C2<br>Image/status     | <b>Out:</b> frame: ImageRecord<br><b>In:</b> candidate_frame: ImageRecord<br>image: digital image; capture_status: text                                                                                                      | Digital image with capture status | Camera image-transfer interface               | Reject unusable images; recapture within the retry limit or request review. | C2 validates and retries.                                 |
| I5  | C2 → C4<br>Hardware command | <b>Out:</b> command: HardwareCommand<br><b>In:</b> hardware_request: HardwareCommand<br>action: enum(pulse, stop, light); duration_ms: nonnegative integer; intensity: real in 0-1; light_enabled: boolean; request_id: text | Structured control message        | Local controller command/status link          | Reject invalid commands; communication failure stops the counting cycle.    | C4 validates and stops; C2 handles communication failure. |
| I6  | C4 → C2<br>Hardware status  | <b>Out:</b> status: HardwareStatus<br><b>In:</b> control_status: HardwareStatus<br>request_id: text; complete: boolean; drive_off: boolean; error: text or none                                                              | Structured status message         | Local controller command/status link          | Missing or failed completion prevents final-count capture.                  | C4 reports; C2 gates capture.                             |
| I7  | C4 → C5<br>Drive control    | <b>Out:</b> drive: MotorControl<br><b>In:</b> motor_control: MotorControl<br>enabled: boolean; intensity: numeric level                                                                                                      | Electrical control signal         | Wired motor-driver connection                 | Disable drive on a detected fault; report failed actuation.                 | C4 stops and reports.                                     |
| I8  | C5 → C6<br>Tray excitation  | <b>Out:</b> excitation: MechanicalMotion<br><b>In:</b> tray_excitation: MechanicalMotion<br>type: vibration                                                                                                                  | Physical vibration                | Mechanical coupling between assembly and tray | Persistent clustering leads to a bounded retry or operator review.          | C2 decides retry/review; C4 stops drive.                  |
| I9  | C7 → C6<br>Diffuse light    | <b>Out:</b> illumination: LightField<br><b>In:</b> incident_light: LightField<br>type: diffuse visible light                                                                                                                 | Optical illumination              | Optical path from lighting to tray            | Poor illumination causes image rejection and review/correction.             | C2 checks image quality.                                  |
| I10 | C6 → C3<br>Reflected scene  | <b>Out:</b> scene: OpticalScene<br><b>In:</b> observed_scene: OpticalScene<br>content: tray and pill appearance                                                                                                              | Reflected optical scene           | Optical path from tray through camera lens    | Unusable visibility leads to recapture or review.                           | C2 evaluates captured images.                             |



| ID  | Connection                    | Named inputs / outputs and types                                                                                                                                                                                                                     | Format                                                                     | Protocol / connection method                                       | On error                                                                                                   | Handled by                                                                           |
|-----|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| I11 | C4 → C7<br>Light control      | <b>Out:</b> command: LightControl<br><b>In:</b> light_command: LightControl<br>enabled: boolean                                                                                                                                                      | Electrical on/off signal                                                   | Wired lighting-control connection                                  | Pause counting on detected control failure; reject poorly lit images.                                      | C4 handles control failure; C2 checks images.                                        |
| I12 | C2 → C8<br>Inference request  | <b>Out:</b> request: InferenceRequest<br><b>In:</b> inference_request: InferenceRequest<br>request_id: text; image: validated RGB image; width, height: positive integers                                                                            | Typed image record locally; image upload plus metadata for an external API | Local adapter call; optional HTTPS request through that adapter    | Reject invalid frames; timeout or unavailable inference yields review/error without an accepted count.     | C8 adapter reports failure; C2 ends or retries the workflow within its budget.       |
| I13 | C8 → C2<br>Inference estimate | <b>Out:</b> estimate: InferenceResult<br><b>In:</b> candidate_estimate: InferenceResult<br>request_id: text; count: nonnegative integer or null; confidence: real in 0–1 or null; status: enum(ok, error); uncertain_regions: array of image regions | Typed result record; external JSON normalized by C8 adapter                | Local adapter return; optional HTTPS response through that adapter | Reject malformed, stale or failed results; missing uncertainty information cannot imply a confident count. | C8 normalizes provider output; C2 validates and chooses acceptance, retry or review. |

Example payload: I2 count result

```
{
  "count": 30,
  "confidence": 0.96,
  "status": "complete"
}
```

This illustrative payload is from Part 4. A result needing review has no accepted count and carries a review status.

# Physical interfaces: tray vibration/pill spreading system

C5/C6 · I7 electrical drive and I8 mechanical excitation

The I5/I6 command-and-status contract controls this mechanism through C4. I7 becomes motor drive at C5; I8 transfers excitation through the carrier to C6. Compliant supports separate tray motion from the stationary camera bridge (D3).

**PillVision · Carrier underside and actuator section.**

**Goal:** Estimate pill count on a camera-visible tray, including touching or partially overlapping pills, and flag uncertain results for operator review.

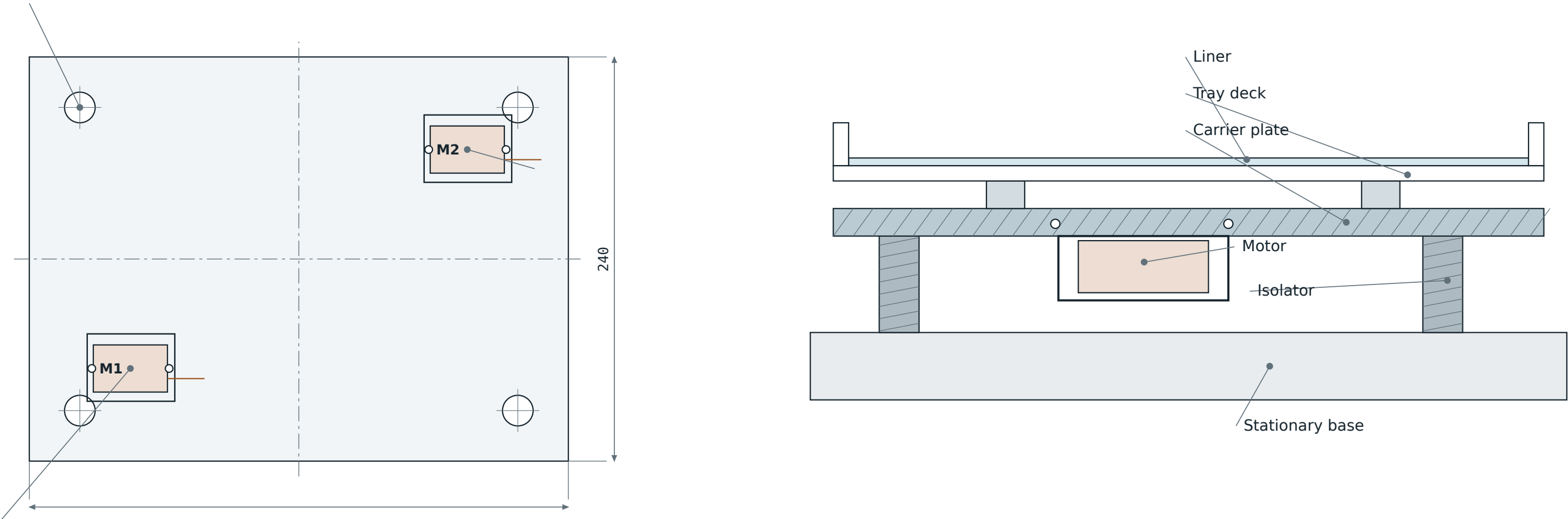


Figure 3: Carrier underside and actuator section.



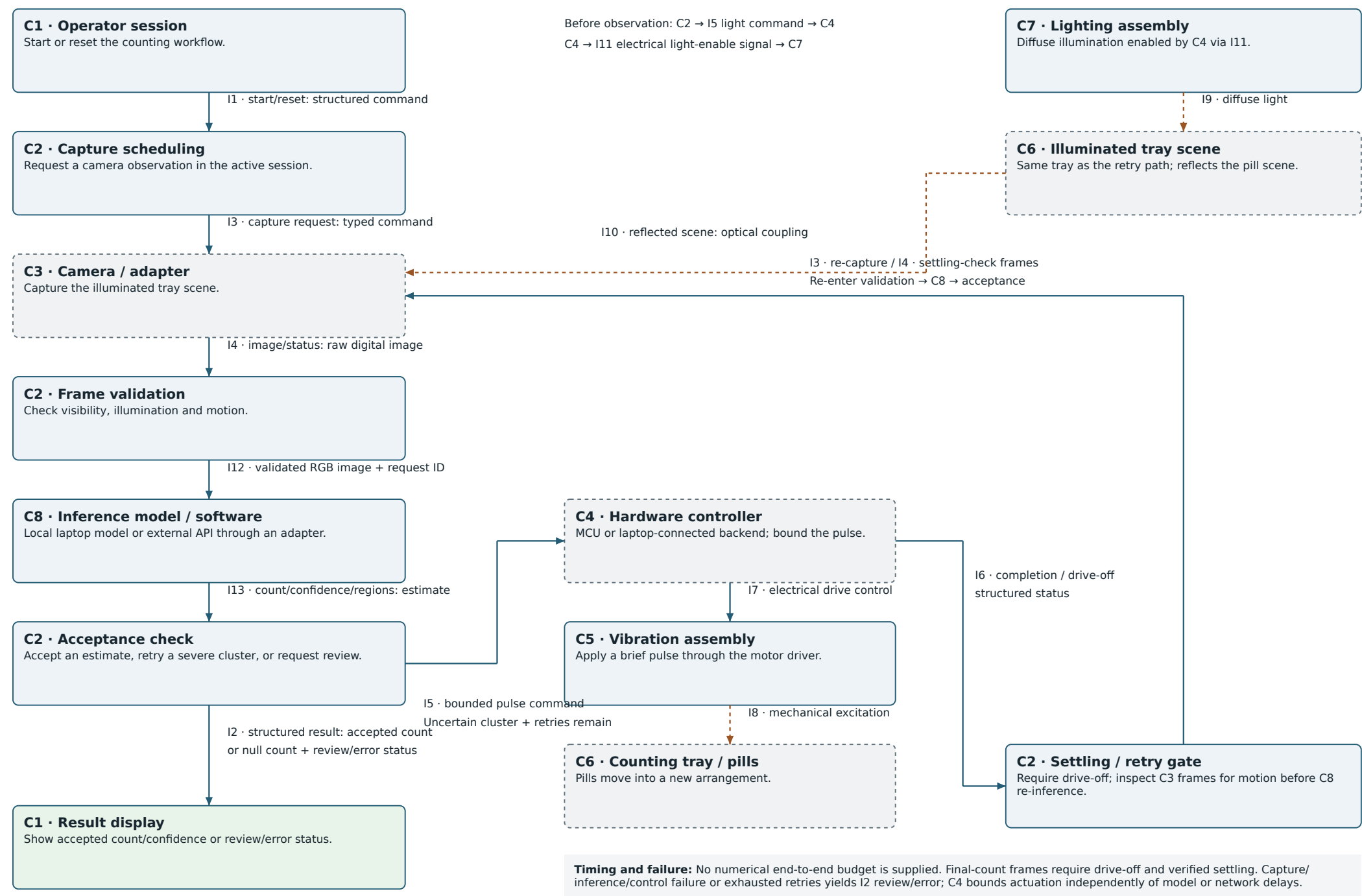
# 5. Data-flow diagrams

Normal counting, low-confidence retry and optical/mechanical paths

The supplied Part 5 flow is integrated below with the updated C1–C8 components and I1–I13 interfaces. C2 schedules capture and validates images; C8 performs model inference; C2 evaluates the returned estimate. Only uncertainty that may benefit from spreading, with retries remaining, triggers vibration. Other failures or exhausted retries request operator review.

PillVision · Normal counting, bounded retry and optical paths.

**Goal:** Estimate pill count on a camera-visible tray, including touching or partially overlapping pills, and flag uncertain results for operator review.



**Legend:** Blue boxes are software processes or team-integrated assemblies; dashed gray boxes are hardware/physical stages. Green marks the operator-facing result. Solid blue arrows carry data/control, including electrical commands; dashed brown arrows show optical/mechanical coupling. C# identifies the responsible component; repeated C2/C6 boxes are stages/views of the same component. I# matches the interface table. C8 performs inference; C2 owns validation, acceptance and retries.

Figure 4: Normal counting, bounded retry and optical paths.

# Data flow: pill spreading and counting sequence

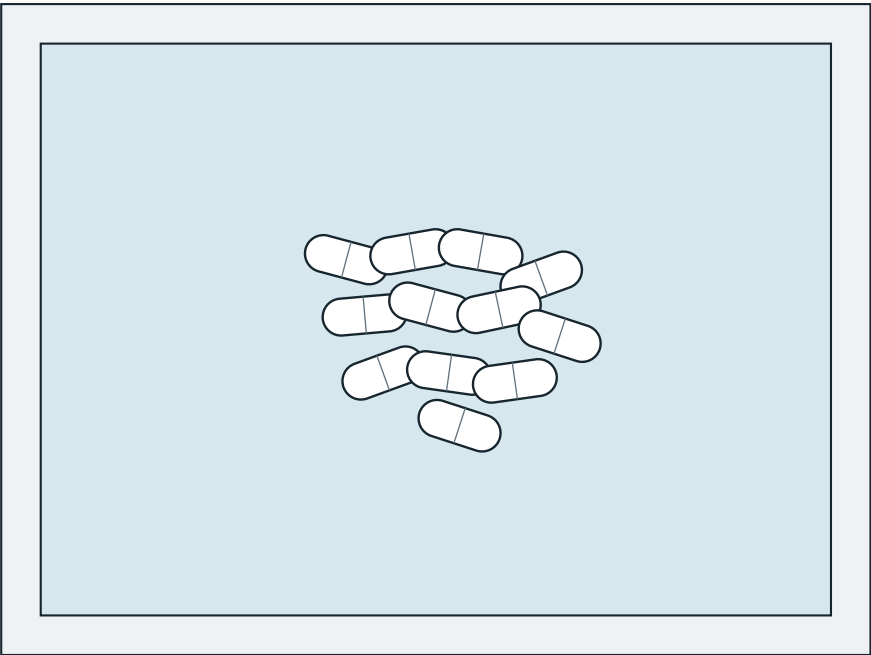
Visual example of observation, pulse and reobservation

These scenes illustrate the preceding data-flow diagrams: I3/I4 acquire an observation, I5-I8 pulse and settle the tray when a retry is warranted, and I2 publishes the accepted count or review status.

**PillVision · Pill spreading and counting sequence.**

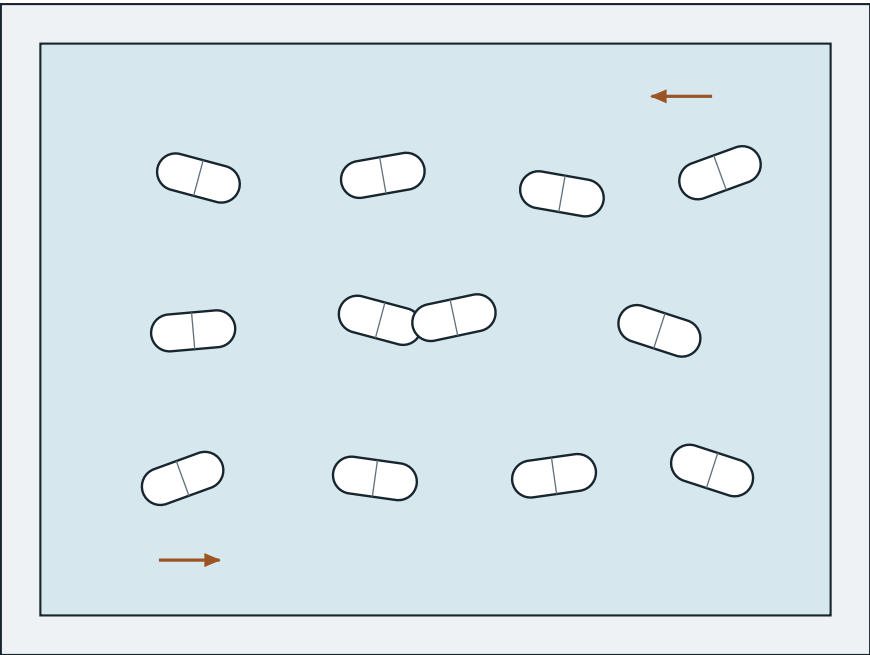
**Goal:** Estimate pill count on a camera-visible tray, including touching or partially overlapping pills, and flag uncertain results for operator review.

**01 · Load and observe**



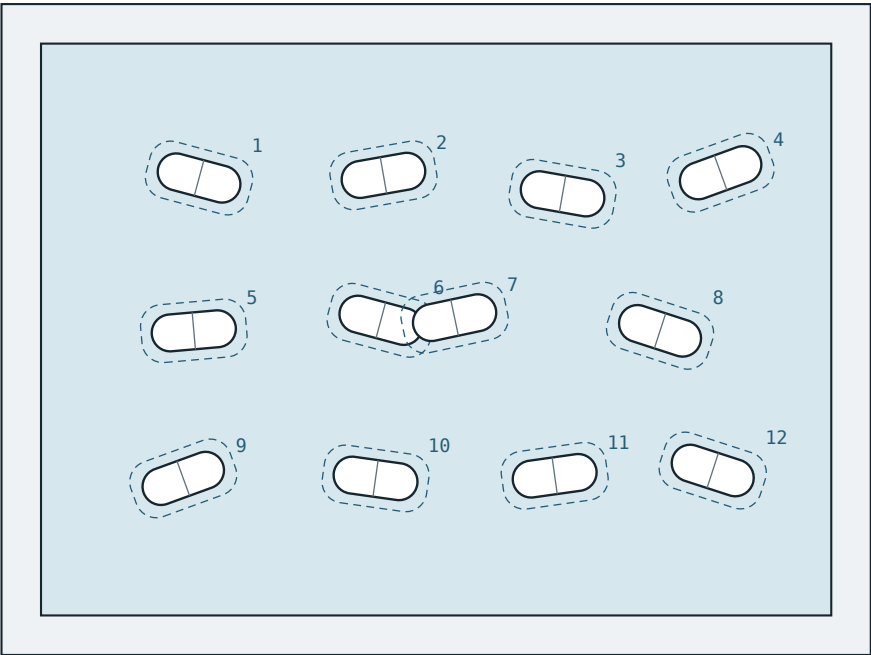
Dense cluster with touching and partially overlapping pills. The system estimates count and flags uncertain regions.

**02 · Pulse and settle**



Apply a brief bounded pulse if needed. Disable motors and wait for settling. Some pills still touch after movement.

**03 · Reimage and review**



Attempt instance detection again and compare observations. Overlays show intended detections, not demonstrated accuracy.

**Legend:** Panels show load/observe, pulse/settle and reimage/review stages. Capsules are pills; brown arrows indicate a pulse direction illustration; Each scene has 12 known illustrated pills; this is not a measured count.

Figure 5: Pill spreading and counting sequence.

## Observation and uncertainty

Each illustrated scene contains 12 pills. That known scene total is not a measured count. The example intentionally retains a touching pair after the pulse to reflect the project goal of counting without complete mechanical separation.

The I2 result includes a count, confidence and status.

## Decision logic

| Observation                         | Next action                     |
|-------------------------------------|---------------------------------|
| Count passes acceptance checks      | Present result for confirmation |
| Severe cluster and retries remain   | Pulse, settle and re-image      |
| Uncertainty persists at retry limit | Request operator review         |

## 6. Architecture pattern and justification

Pipeline plus embedded sensor-actuator control, with optional client-server inference

The **pipeline** governs C3 capture → C2 validation → C8 inference → C2 acceptance → C1 display. The **embedded sensor-actuator** pattern governs camera feedback and C2/C4 motor and lighting control. A **layered** application separates C1 presentation, C2 workflow and C3/C4/C8 adapters. If C8 uses an external inference API, **client-server** governs that request/response boundary only. Local inference uses an adapter call without a remote server.

| Criterion              | Why this combination fits PillVision                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fit to the problem     | Counting is a sequence of image transformations, while spreading requires a feedback loop. Keeping those steps as two flows prevents moving frames from being treated as final observations.                                                                                                                                                                                                                                                       |
| Team skills            | The three-member team can share implementation and testing without assuming permanent roles.                                                                                                                                                                                                                                                                                                                                                       |
| Performance and timing | Local laptop inference avoids uploading frames; external inference adds upload and service latency, so requests require bounded waits. C4 bounds pulse duration independently of inference; asynchronous capture/control waits keep C1 responsive. Capture and settling gates address the ordering requirement before throughput is tuned.                                                                                                         |
| Scalability            | The immediate target is one local tray/session, not a multi-tenant service. Additional devices can run independent local sessions. A second camera can later enter through another C3 adapter without changing the result contract to C1.                                                                                                                                                                                                          |
| Hardware constraints   | The fixed camera observes a moving tray, it can only vibrate the tray to spread the pills so it can't handle a case where 2 are stuck together. A laptop can provide memory/compute for C8 while an MCU handles actuation. A laptop-hosted controller still needs a driver interface with independent pulse cutoff. MCU-led external inference requires feasible connectivity and a compatible provider; device capabilities remain to be checked. |

### Rejected alternative

This proposed architecture rejects splitting each local counting stage into independently deployed microservices: separate network services would add network failure handling and image-transfer overhead to a single local device without addressing overlap or physical settling. An optional external inference service is a specific client-server dependency, not a requirement to distribute every stage. The UI and processor remain separate logical components inside one local application. This is a proposed design and it is likely to be modified or expanded.

# 7. Decision log

Design choices, alternatives and rationale

| ID | Proposed decision                                                                                  | Alternative options                                                                                               | Rationale for this option                                                                                                                                                                                                                                                                                                       | Affected components/<br>interfaces            |
|----|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| D1 | Use one fixed overhead camera for the first version.                                               | Side camera; overhead camera; multiple cameras.                                                                   | An overhead camera gives a clear view of the full counting tray and keeps the first version simpler. A second view remains an option if later tests show a useful improvement.                                                                                                                                                  | C2, C3; I3, I4, I10                           |
| D2 | Mount the camera on the stationary bridge and the motors on the isolated carrier.                  | Camera mounted on the moving tray; rigidly coupling the tray to the camera structure.                             | Separate structural paths reduce disturbance at the camera while allowing controlled tray motion. Cable slack prevents wiring from bypassing isolation.                                                                                                                                                                         | C3, C5, C6; I8, I10                           |
| D3 | Separate C8 inference from C2 coordination; prefer laptop inference with bounded hardware control. | External inference API; MCU-only inference if feasible; laptop-hosted control backend; distributed microservices. | A replaceable inference adapter supports local or external models. Laptop compute avoids assuming the MCU can fit the model. Independent control bounds keep model or network delays from extending actuation; final deployment remains proposed.                                                                               | C1–C8; I1–I13                                 |
| D4 | Use camera-based inference as the primary counting method.                                         | Motion sensing; laser/beam counting of individually separated pills; camera plus supplementary sensors.           | Camera images preserve pill shape and spatial context for C8 to estimate touching or partially overlapping instances. Motion sensing alone does not resolve pill count; beam counting would need a controlled passage and separation mechanism. Vibration remains an optional aid, with uncertain observations sent for review. | C2, C3, C6, C7, C8; I3, I4, I9, I10, I12, I13 |
| D5 | Use vibration only when needed.                                                                    | No vibration; always vibrate; vibrate only when the count is uncertain.                                           | Short vibration pulses can spread severe clusters without adding unnecessary movement to every count. C2 requests a bounded pulse only when an uncertain observation may benefit from spreading and retries remain; the system then checks settling before re-imaging. Complete pill separation is not required.                | C2, C4, C5, C6; I5, I6, I7, I8                |